Switching Circuits Laboratory (CS29002)

<u>Assignment 9 (01/08-04-2026)</u>

a) Design a circuit that would generate successive 4-bit Fibonacci numbers in response to a manual clock. The generated sequence will be: 0, 1, 1, 2, 3, 5, 8, 13, after which overflow occurs. There is a one-bit input called **START**, which is normally **0**; to start the process, we have to set **START = 1**. At the next clock, the necessary initialization will be carried out, and the Fibonacci numbers will be generated on successive clocks.

   i) Draw the circuit for the data path, which would contain registers, an adder, and any other circuits as required. The required control signals must be shown clearly.

   ii) Draw the state transition diagram for the control path, which will generate the required control signals for the data path, in synchronism with the applied clock. Implement the control path using J-K flip-flops and verify the circuit's operation.

b) It is required to design a circuit that can display four different digital waveforms on a single oscilloscope channel. The circuit will contain a 4-to-1 multiplexer to select one channel at a time, a 2-bit counter to drive the multiplexer's select lines, a D/A converter to generate a voltage offset proportional to the counter output, and a voltage-addition circuit using an operational amplifier.

   Draw the complete circuit diagram and implement it using ICs as required. Test the circuit using the waveforms generated at the outputs of a 4-bit counter.

<u>Instructions:</u>

1) *Clearly show the complete design for both the data path and the control path.*
2) *Draw the circuit diagram on paper, following the suggested conventions. Following this, you can issue the components and start with the realization on the breadboard.*
3) *Prepare a laboratory report in PDF format and upload it to Moodle by the specified deadline. A single report has to be uploaded per group.*